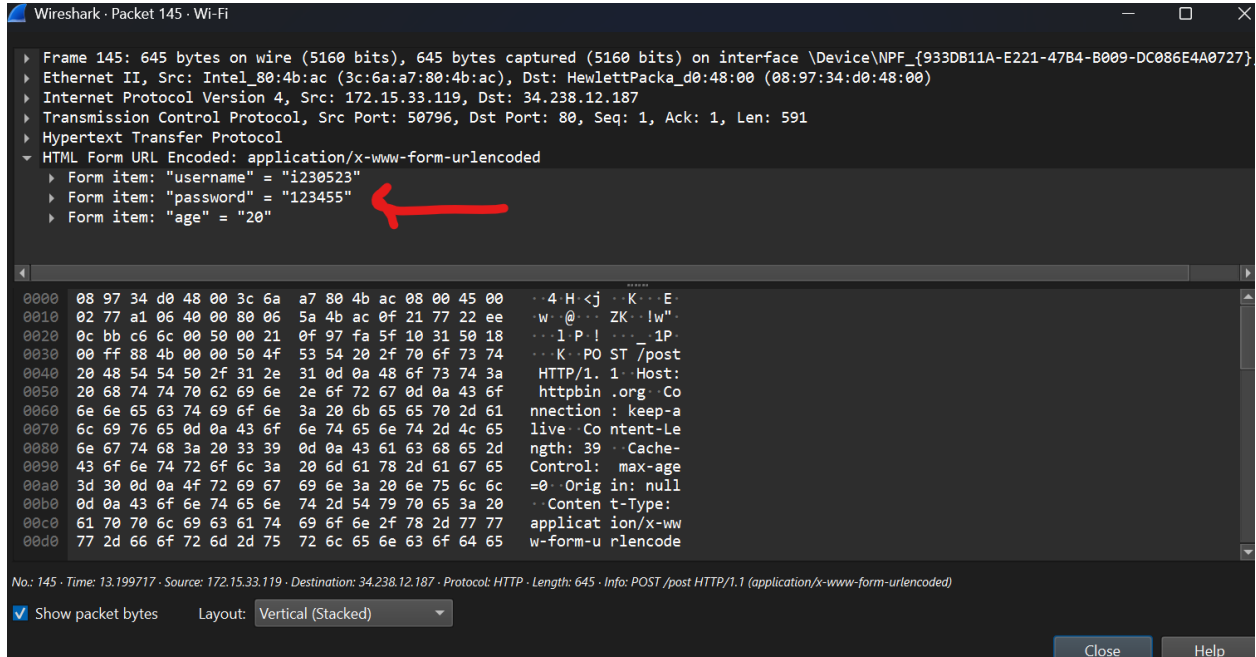


CNET Lab Task 3

Task 1:



The image shows a Wireshark packet capture window for 'Wi-Fi'. The selected packet is 145, which is an HTTP POST request. The packet details pane shows the following structure:

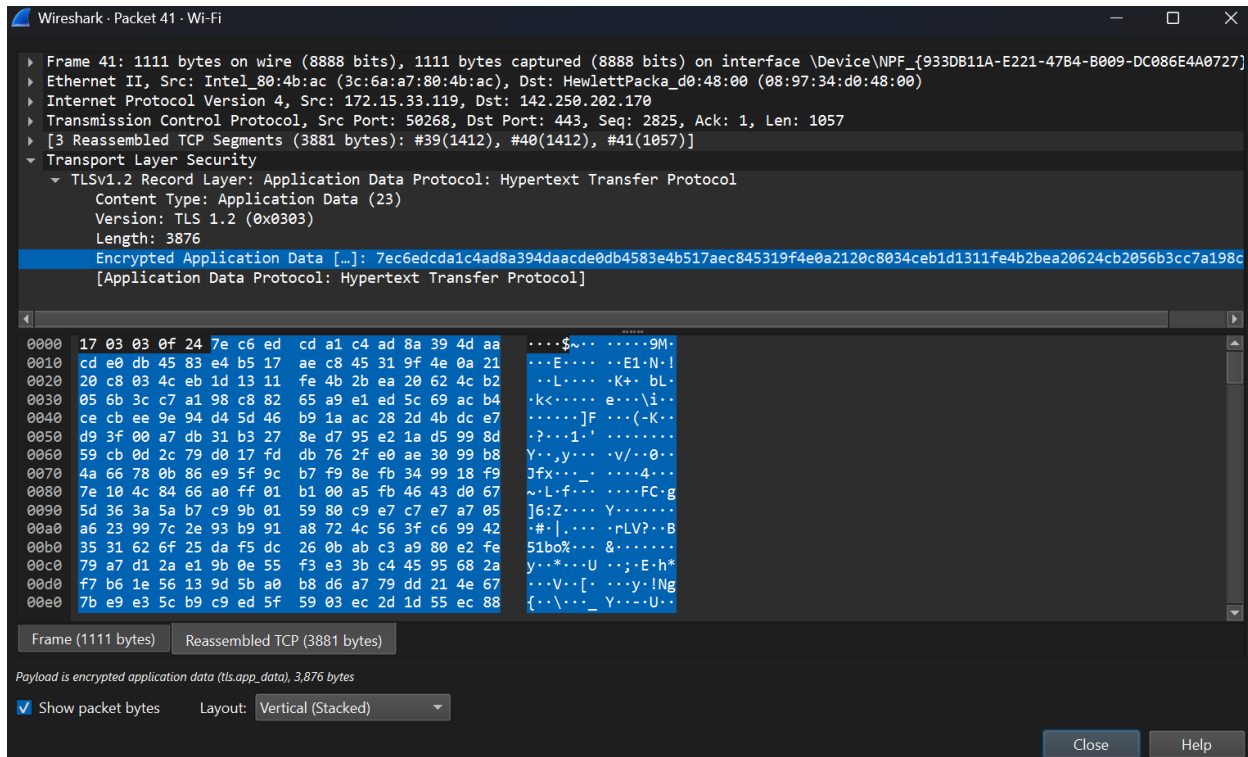
- Frame 145: 645 bytes on wire (5160 bits), 645 bytes captured (5160 bits) on interface \Device\NPF_{933DB11A-E221-47B4-B009-DC086E4A0727}
- Ethernet II, Src: Intel_80:4b:ac (3c:6a:a7:80:4b:ac), Dst: HewlettPacka_d0:48:00 (08:97:34:d0:48:00)
- Internet Protocol Version 4, Src: 172.15.33.119, Dst: 34.238.12.187
- Transmission Control Protocol, Src Port: 50796, Dst Port: 80, Seq: 1, Ack: 1, Len: 591
- Hypertext Transfer Protocol
- HTML Form URL Encoded: application/x-www-form-urlencoded
 - Form item: "username" = "i230523"
 - Form item: "password" = "123455" (highlighted with a red arrow)
 - Form item: "age" = "20"

The packet bytes pane shows the raw data of the packet, which is the URL-encoded form data. The status bar at the bottom indicates: No.: 145 - Time: 13.199717 - Source: 172.15.33.119 - Destination: 34.238.12.187 - Protocol: HTTP - Length: 645 - Info: POST /post HTTP/1.1 (application/x-www-form-urlencoded).

Q1. Why is HTTP considered insecure for transmitting sensitive data? Explain the risks of plaintext transmission and how attackers can exploit it.

Ans: HTTP is insecure bcz everything in it is not encrypted so any sensitive data like personal info like password, etc can be seen in text form by anyone. So attackers will be able to see the password easily.

Task 2:



Q1. What differences do you notice between HTTP and HTTPS traffic? Compare the packet details for HTTP vs. HTTPS and note how the data is secured.

Ans: In HTTP, the data of application is all shown in plain text where as in HTTPS, the data is shown all in encrypted form which we can not read.

Q2. Why is HTTPS considered more secure than HTTP? Explain the role of SSL/TLS encryption in preventing data interception and tampering.

Ans: HTTPS is considered more secure than HTTP bcz data in https is encrypted and in http, it is not. The SSL/TLS encrypts the connection between client and server which prevents tampering and interception of data during data transmission.

Task 3:

1. Packet arrival time (timestamp) Arrival Time: Sep 4, 2025 12:39:17.975625000
Pakistan Standard Time

2. Frame length (bytes) Frame Length: 74 bytes (592 bits)

3. Time difference between consecutive packets (inter-packet delay) [first and last request] 12:39:22.032186000 - 12:39:17.975625000 = 4.056561

Q1. How frequently are packets being sent?

Ans: Every 1.02 seconds

Q2. What is the average packet size, and how does it compare to other traffic types? (hint: you will have to generate different traffic)

Ans: Avg packet size of HTTPS is 74 bytes meanwhile of HTTP, it is more than 500 bytes.

Q3. What transmission mode is used?

Ans: N/A

Task 4:

```
C:\Users\Dell>nslookup wikipedia.org
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     wikipedia.org
Addresses: 2001:df2:e500:ed1a::1
          103.102.166.224
```

No.	Time	Source	Destination	Protocol	Length	Info
413	22.790484	172.15.33.119	8.8.8.8	DNS	74	Standard query 0x3c5c A www.google.com
414	22.790685	172.15.33.119	8.8.8.8	DNS	74	Standard query 0x78df HTTPS www.google.com
415	22.827228	8.8.8.8	172.15.33.119	DNS	99	Standard query response 0x78df HTTPS www.google.com HTTPS
416	22.841414	8.8.8.8	172.15.33.119	DNS	90	Standard query response 0x3c5c A www.google.com A 142.250.200.164
447	24.717743	172.15.33.119	8.8.8.8	DNS	79	Standard query 0xb95d A accounts.google.com
448	24.718368	172.15.33.119	8.8.8.8	DNS	79	Standard query 0x4b67 HTTPS accounts.google.com
449	24.768884	8.8.8.8	172.15.33.119	DNS	95	Standard query response 0xb95d A accounts.google.com A 142.251.12.84
451	24.854128	8.8.8.8	172.15.33.119	DNS	129	Standard query response 0x4b67 HTTPS accounts.google.com SOA ns1.google.com
603	29.307195	172.15.33.119	8.8.8.8	DNS	80	Standard query 0x0001 PTR 8.8.8.8.in-addr.arpa
604	29.346561	8.8.8.8	172.15.33.119	DNS	104	Standard query response 0x0001 PTR 8.8.8.8.in-addr.arpa PTR dns.google
605	29.349012	172.15.33.119	8.8.8.8	DNS	73	Standard query 0x0002 A wikipedia.org
606	29.565977	8.8.8.8	172.15.33.119	DNS	89	Standard query response 0x0002 A wikipedia.org A 103.102.166.224
607	29.573016	172.15.33.119	8.8.8.8	DNS	73	Standard query 0x0003 AAAA wikipedia.org
608	29.706274	8.8.8.8	172.15.33.119	DNS	101	Standard query response 0x0003 AAAA wikipedia.org AAAA 2001:df2:e500:edia::1
1036	47.168514	172.15.33.119	8.8.8.8	DNS	76	Standard query 0x5af9 A drive.google.com

Frame 605: 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface \Dev	0000	00 97 34 d0 48 00 3c 6a	a7 80 4b ac 08 00 45 00	· 4 H < j · K · E ·
Ethernet II, Src: Intel_80:4b:ac (3c:6a:a7:80:4b:ac), Dst: HewlettPacka_d0:48:00 (08:9	0010	00 3b 8a dc 00 00 00 11	d2 3f ac 0f 21 77 08 08	; · · · · · ? · lw ·
Internet Protocol Version 4, Src: 172.15.33.119, Dst: 8.8.8.8	0020	00 08 c6 17 00 35 00 27	37 cb 00 02 01 00 00 01	· · · · · S ' 7 · · · · ·
User Datagram Protocol, Src Port: 50711, Dst Port: 53	0030	00 00 00 00 00 00 09 77	69 6b 69 70 65 64 69 61	· · · · · w ipedia
Domain Name System (query)	0040	03 6f 72 67 00 00 01 00	01	org · · · ·


```

Transaction ID: 0x0002
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
Queries
[Response In: 606]

```

DNS Response Packet

Q1: Why is DNS critical for HTTP/HTTPS communication, and how does it relate to secure data transmission?

Ans: DNS is critical bcz it translates human readable domain names into computer readable domain names, to enable HTTP/HTTPS to reach connection to reach to connect the server successfully, and it does not encypt the ip addresses.